ENGT 320 Applied Thermal Systems
Quiz 4 Formula Sheet

$$m = \rho \forall$$
$$\dot{\forall} = \frac{\forall}{t}$$
$$q = mc_p \Delta T$$
$$q = -kA\frac{dT}{dt}$$
$$q = hA\left(T_s - T_\infty\right)$$
$$q = \varepsilon \sigma A\left(T_s^4 - T_{sur}^4\right)$$
$$\sigma = 5.67 \times 10^{-8} \frac{W}{m^2 K^4}$$
$$\Delta E_{sys} = \Sigma E_{in} - \Sigma E_{out}$$
$$\frac{dE_{sys}}{dt} = \Sigma \dot{E}_{in} - \Sigma \dot{E}_{out}$$
$$q'' = \frac{q}{A}$$
$$\hat{q}'' = -k_x \frac{\partial T}{\partial x}\hat{i} - k_y \frac{\partial T}{\partial y}\hat{j} - k_z \frac{\partial T}{\partial z}\hat{k}$$
$$k = \frac{9\gamma - 5}{4} \frac{c_v}{\pi d^2} \sqrt{\frac{M_w k_B T}{N_A \pi}}$$
$$k_B = 1.381 \times 10^{-23} \frac{J}{K}$$
$$N_A = 6.022 \times 10^{23}$$
$$\alpha = \frac{k}{\rho c_p}$$
$$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx$$
$$q_{y+dy} = q_y + \frac{\partial q_y}{\partial y} dy$$
$$q_{z+dz} = q_z + \frac{\partial q_z}{\partial z} dz$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = 0$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial \phi^2} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial T}{\partial \theta}\right) + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = 0$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$T(0,t) = T_s$$
$$-k\frac{\partial T}{\partial x}\Big|_{x=0} = q_s''$$
$$\frac{\partial T}{\partial x}\Big|_{x=0} = 0$$
$$-k\frac{\partial T}{\partial x}\Big|_{x=0} = h\left(T_\infty - T(0,t)\right)$$

$$C_1 = \frac{T_{s,2} - T_{s,1}}{L}$$
$$C_2 = T_{s,1}$$
$$T(x) = \left(T_{s,2} - T_{s,1}\right)\frac{x}{L} + T_{s,1}$$
$$q = \frac{kA}{L}\left(T_{s,1} - T_{s,2}\right)$$
$$R_{t,cond} = \frac{T_{s,1} - T_{s,2}}{q} = \frac{L}{kA}$$
$$R = \frac{V}{I}$$
$$R_{t,conv} = \frac{T_s - T_\infty}{q} = \frac{1}{hA}$$
$$R_{t,rad} = \frac{T_s - T_{sur}}{q} = \frac{1}{h_r A}$$
$$h_r = \varepsilon\sigma\left(T_s + T_{sur}\right)\left(T_s^2 + T_{sur}^2\right)$$
$$q = UA\Delta T$$
$$U = \frac{1}{R_{tot}A}$$
$$R_{tot} = \Sigma R_t = \frac{1}{UA}$$

$$R_{t,c}'' = \frac{T_A - T_B}{q''}$$
$$q = \frac{k_{eff}A}{L}\left(T_1 - T_2\right)$$
$$k_{eff,min} = \frac{1}{\frac{1-\varepsilon}{k_s} + \frac{\varepsilon}{k_f}}$$
$$k_{eff,max} = \varepsilon k_f + (1-\varepsilon)k_s$$
$$R_{t,cond} = \frac{\ln\frac{r_2}{r_1}}{2\pi kL}$$
$$R_{t,conv} = \frac{1}{h2\pi rL}$$

$$\frac{d^2T}{dx^2} + \left(\frac{1}{A_c}\frac{dA_c}{dx}\right)\frac{dT}{dx} - \left(\frac{1}{A_c}\frac{h}{k}\frac{dA_s}{dx}\right)\left(T - T_\infty\right) = 0$$

$$A_s = Px$$
$$\frac{d^2T}{dx^2} - \left(\frac{hP}{kA_c}\right)\left(T - T_\infty\right)$$
$$\frac{d^2\theta}{dx^2} - m^2\theta = 0$$
$$\theta = T(x) - T_\infty$$
$$m^2 = \frac{hP}{kA_c}$$
$$P = 2w + 2t$$
$$A_c = wt$$
$$P = \pi D$$
$$A_c = \frac{\pi}{4}D^2$$
$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$
$$\theta_b = T_b - T_\infty$$
$$M = \sqrt{hPKA_c}\theta_b$$
$$\varepsilon_f = \frac{q_f}{hA_{c,b}\theta_b}$$
$$\varepsilon_f = \sqrt{\frac{kP}{hA_c}}$$
$$R_{t,f} = \frac{\theta_b}{q_f}$$
$$R_{t,b} = \frac{1}{hA_{c,b}}$$
$$\varepsilon_f = \frac{R_{t,b}}{R_{t,f}}$$
$$\eta_f = \frac{q_f}{q_{max}} = \frac{q_f}{hA_f\theta_b}$$
$$\eta_o = \frac{q_t}{q_{max}} = \frac{q_t}{hA_t\theta_b}$$
$$A_t = NA_f + A_b$$
$$q_t = N\eta_f hA_f\theta_b + hA_b\theta_b$$
$$\eta_o = 1 - \frac{NA_f}{A_t}\left(1 - \eta_f\right)$$
$$R_{t,o} = \frac{\theta_b}{q_t} = \frac{1}{\eta_o hA_t}$$

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$$\frac{\partial^2 T}{\partial x^2}\Big|_{m,n} \approx \frac{\frac{\partial T}{\partial x}\Big|_{m+\frac{1}{2},n} - \frac{\partial T}{\partial x}\Big|_{m-\frac{1}{2},n}}{\Delta x}$$

$$\frac{\partial T}{\partial x}\Big|_{m+\frac{1}{2},n} \approx \frac{T\Big|_{m+1,n} - T\Big|_{m,n}}{\Delta x}$$

$$\frac{\partial T}{\partial x}\Big|_{m-\frac{1}{2},n} \approx \frac{T\Big|_{m,n} - T\Big|_{m-1,n}}{\Delta x}$$

$$\frac{\partial^2 T}{\partial x^2}\Big|_{m,n} \approx \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{(\Delta x)^2}$$

$$\frac{\partial^2 T}{\partial y^2}\Big|_{m,n} \approx \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{(\Delta y)^2}$$

$$\sum_{i=1}^{4} q_{(i)\to(m,n)} + \dot{q}\,(\Delta x \cdot \Delta y \cdot 1) = 0$$

$$q_{(m-1,n)\to(m,n)} = k\,(\Delta y \cdot 1)\,\frac{T_{m-1,n} - T_{m,n}}{\Delta x}$$

$$q_{(m+1,n)\to(m,n)} = k\,(\Delta y \cdot 1)\,\frac{T_{m+1,n} - T_{m,n}}{\Delta x}$$

$$q_{(m,n+1)\to(m,n)} = k\,(\Delta x \cdot 1)\,\frac{T_{m,n+1} - T_{m,n}}{\Delta y}$$

$$q_{(m,n-1)\to(m,n)} = k\,(\Delta x \cdot 1)\,\frac{T_{m,n-1} - T_{m,n}}{\Delta y}$$

$$T_{m,n+1} + T_{m,n-1} + T_{m+1,n} + T_{m-1,n} + \frac{\dot{q}\cdot 1\cdot(\Delta x)^2}{k} - 4T_{m,n} = 0$$

$$AT = C$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{bmatrix}$$

$$T = \begin{bmatrix} T_1 \\ T_2 \\ \vdots \\ T_N \end{bmatrix}$$

$$C = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_N \end{bmatrix}$$

$$T = A^{-1}C$$

$$\frac{u(y)}{u_\infty} = 0.99$$

$$C_f = \frac{\tau_s}{\rho u_\infty^2/2}$$

$$\tau_s = \mu \frac{\partial u}{\partial y}\Big|_{y=0}$$

$$\frac{T_s - T(y)}{T_s - T_\infty} = 0.99$$

$$q_s'' = -k\frac{\partial T}{\partial y}\Big|_{y=0}$$

$$q_s'' = h\,(T_s - T_\infty)$$

$$h = \frac{-k\frac{\partial T}{\partial y}\Big|_{y=0}}{T_s - T_\infty}$$

$$q = \int_{A_s} q''\,dA_s = (T_s - T_\infty)\int_{A_s} h\,dA_s$$

$$\overline{q}'' = \overline{h}\,(T_s - T_\infty)$$

$$\overline{q} = \overline{q}''\,A_s = \overline{h}A_s\,(T_s - T_\infty)$$

$$\overline{h} = \frac{1}{A_s}\int_{A_s} h\,dA_s$$

$$\overline{h} = \frac{1}{L}\int_0^L h\,dx$$

$$Re_x = \frac{\rho u_\infty x}{\mu}$$

$$x^* = \frac{x}{L}$$

$$y^* = \frac{y}{L}$$

$$u^* = \frac{u}{V}$$

$$\mathrm{v}^* = \frac{\mathrm{V}}{V}$$

$$T^* = \frac{T - T_s}{T_\infty - T_s}$$

$$p^* = \frac{p_\infty}{\rho V^2}$$

$$u^*\frac{\partial u^*}{\partial x^*} + \mathrm{v}^*\frac{\partial u^*}{\partial y^*} = -\frac{dp^*}{dx^*} + \frac{1}{Re_L}\frac{\partial^2 u^*}{\partial y^{*2}}$$

$$Re_L = \frac{VL}{\nu}$$

$$\nu = \frac{\mu}{\rho}$$

$$u^*\frac{\partial T^*}{\partial x^*} + \mathrm{v}^*\frac{\partial T^*}{\partial y^*} = \frac{1}{Re_L Pr}\frac{\partial^2 T^*}{\partial y^{*2}}$$

$$Pr = \frac{\nu}{\alpha}$$

$$Nu = \frac{hL}{k} = \frac{\partial T^*}{\partial y^*}\Big|_{y^*=0}$$

$$\overline{Nu} = \frac{\overline{h}L}{k}$$

$$T_f = \frac{T_s + T_\infty}{2}$$

. . . . . .

Isothermal Flat Plate

$$\frac{\delta}{\delta_t} \approx Pr^{1/3}$$

Laminar:

$$\delta = \frac{5}{\sqrt{\frac{u_\infty}{\nu x}}} = \frac{5x}{\sqrt{Re_x}}$$

$$C_{f,x} = \frac{\tau_{s,x}}{\frac{\rho u_\infty^2}{2}} = 0.664 Re_x^{-1/2}$$

$$Nu_x = \frac{h_x x}{k} = 0.332 Re_x^{1/2} Pr^{1/3}$$

$$(Pr \gtrsim 0.6)$$

$$\overline{C}_{f,x} = 1.328 Re_x^{-1/2}$$

$$\overline{Nu}_x = \frac{\overline{h}_x x}{k} = 0.664 Re_x^{1/2} Pr^{1/3}$$

$$(Pr \gtrsim 0.6)$$

Turbulent:

$$\delta = 0.37 x Re_x^{-1/2}$$

$$C_{f,x} = 0.0592 Re_x^{-1/5}$$

$$\left(Re_{x,c} \lesssim Re_x \lesssim 10^8\right)$$

$$Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3}$$

$$(0.6 \lesssim Pr \lesssim 60)$$

$$A = 0.037 Re_{x,c}^{4/5} - 0.664 Re_{x,c}^{1/2}$$

$$\overline{C}_{f,L} = 0.074 Re_L^{-1/5} - \frac{2A}{Re_L}$$

$$\left(Re_{x,c} \lesssim Re_L \lesssim 10^8\right)$$

$$\overline{Nu}_L = \left(0.037 Re_L^{4/5} - A\right) Pr^{1/3}$$

$$(0.6 \lesssim Pr \lesssim 60)$$

$$\left(Re_{x,c} \lesssim Re_L \lesssim 10^8\right)$$

. . . . . .

$$Re_D = \frac{\rho V D}{\mu} = \frac{VD}{\nu}$$

$$C_D = \frac{F_D}{A_f\left(\frac{\rho V^2}{2}\right)}$$

. . . . . .

Cylinder

$$\overline{Nu}_D = \frac{\overline{h}D}{k} = C Re_D^m Pr^{1/3}$$

$$(Pr \gtrsim 0.7)$$

See Table 5

. . . . . .

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Various Geometries

$\overline{Nu}_D = \frac{\overline{h}D}{k} = CRe_D^m Pr^{1/3}$

$(Pr \gtrsim 0.7)$

See Table 6

.....

.....

Sphere

$A_s = \pi D^2$

$C_D = \frac{25}{Re_D}$

$(Re_D \lesssim 0.5)$

$\overline{Nu}_D = 2 + \left(0.4Re_D^{1/2} + 0.06Re_D^{2/3}\right) Pr^{0.4} \left(\frac{\mu}{\mu_s}\right)^{1/4}$

$(0.71 \lesssim Pr \lesssim 380)$

$\left(3.5 \lesssim Re_D \lesssim 7.6 \times 10^4\right)$

$\left(1.0 \lesssim \frac{\mu}{\mu_s} \lesssim 3.2\right)$

.....

$m = \rho u_m A_c$

$Re_D = \frac{\rho u_m D}{\mu} = \frac{u_m D}{\nu}$

$\left(\frac{x_{fd,t}}{D}\right) \approx 0.05 Re_D Pr$

$f = \frac{-\frac{dP}{dx} D}{\rho \frac{u_m^2}{2}}$

$f = \frac{64}{Re_D}$

$\frac{1}{\sqrt{f}} = -2.0 \log\left[\frac{e/D}{3.7} + \frac{2.51}{Re_D\sqrt{f}}\right]$

$dq_{conv} = mc_p\left[(T_m + dT_m) - T_m\right] = mc_p dT_m$

$dq_{conv} = q_s^{''} P dx$

$\frac{dT_m}{dx} = \frac{q_s^{''} P}{mc_p} = \frac{P}{mc_p} h\left(T_s - T_m\right)$

$T_m\left(x\right) = T_{m,i} + \frac{q_s^{''} P}{mc_p} x$

$\Delta T = T_s - T_m$

$\frac{dT_m}{dx} = -\frac{d(\Delta T)}{dx} = \frac{P}{mc_p} h \Delta T$

$\frac{T_s - T_m(x)}{T_s - T_{m,i}} = e^{-\frac{Px}{mc_p}\overline{h}}$

$\Delta T_{lm} = \frac{\Delta T_o - \Delta T_i}{\ln\left(\frac{\Delta T_o}{\Delta T_i}\right)}$

$\Delta T_o = T_s - T_{m,o}$

$\Delta T_i = T_s - T_{m,i}$

$q_{conv} = \overline{h} A_s \Delta T_{lm}$

.....

Circular Tubes

Fully dev. lam. w/ $q^{''} = \mathbb{C}$:

$Nu_D = 4.36$

Fully dev. lam. w/ $T_s = \mathbb{C}$:

$Nu_D = 3.66$

Fully dev. turb. w/ small ΔT:

$Nu_D = 0.023 Re_D^{4/5} Pr^n$

$n = 0.4, \quad T_s > T_m$

$n = 0.3, \quad T_s < T_m$

$(0.6 \lesssim Pr \lesssim 160)$

$(Re_D \gtrsim 10,000)$

$(L/D \gtrsim 10)$

Fully dev. turb. w/ large ΔT:

$Nu_D = 0.027 Re_D^{4/5} Pr^{1/3} \left(\frac{\mu}{\mu_s}\right)^{0.14}$

$(0.7 \lesssim Pr \lesssim 16,700)$

$(Re_D \gtrsim 10,000)$

$(L/D \gtrsim 10)$

.....

.....

Noncircular Tubes

$D_h = \frac{4A_c}{P}$

$Nu_D = \frac{hD_h}{k}$

See Table 7

.....

$\frac{1}{UA} = \frac{1}{U_c A_c} = \frac{1}{U_h A_h} = \frac{1}{(hA)_c} + R_w + \frac{1}{(hA)_h}$

$\frac{1}{UA} = \frac{1}{(\eta_o hA)_c} + \frac{R_{f,c}^{''}}{(\eta_o A)_c} + R_w + \frac{R_{f,h}^{''}}{(\eta_o A)_h} + \frac{1}{(\eta_o hA)_h}$

$q = m_h\left(i_{h,i} - i_{h,o}\right)$

$q = m_c\left(i_{c,i} - i_{c,o}\right)$

$q = m_h c_{p,h}\left(T_{h,i} - T_{h,o}\right)$

$q = m_c c_{p,c}\left(T_{c,o} - T_{c,i}\right)$

$q = UA\Delta T_m$

$q = UA\Delta T_{lm}$

$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\frac{\Delta T_2}{\Delta T_1}} = \frac{\Delta T_1 - \Delta T_2}{\ln\frac{\Delta T_1}{\Delta T_2}}$

$\Delta T_1 = T_{h,1} - T_{c,1} = T_{h,i} - T_{c,i}$

$\Delta T_2 = T_{h,2} - T_{c,2} = T_{h,o} - T_{c,o}$

$\Delta T_1 = T_{h,1} - T_{c,1} = T_{h,i} - T_{c,o}$

$\Delta T_2 = T_{h,2} - T_{c,2} = T_{h,o} - T_{c,i}$

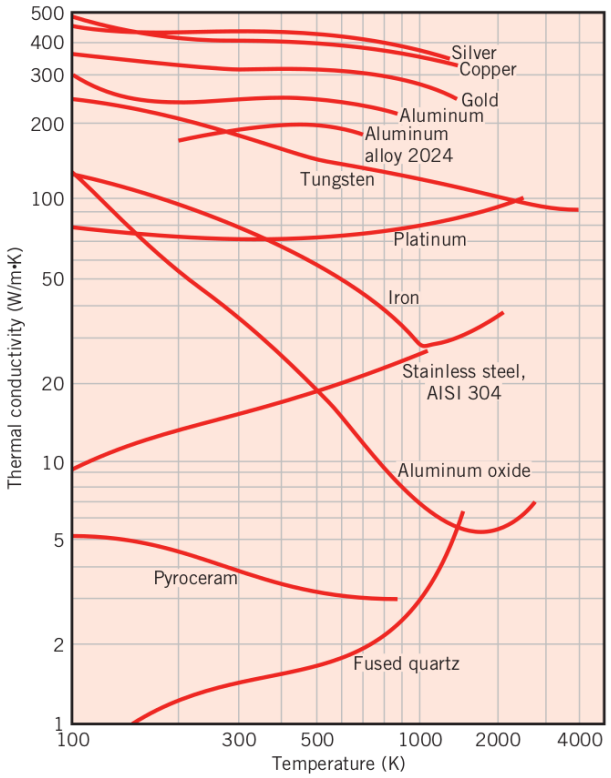
$q_{max} = C_{min}\left(T_{h,i} - T_{c,i}\right)$

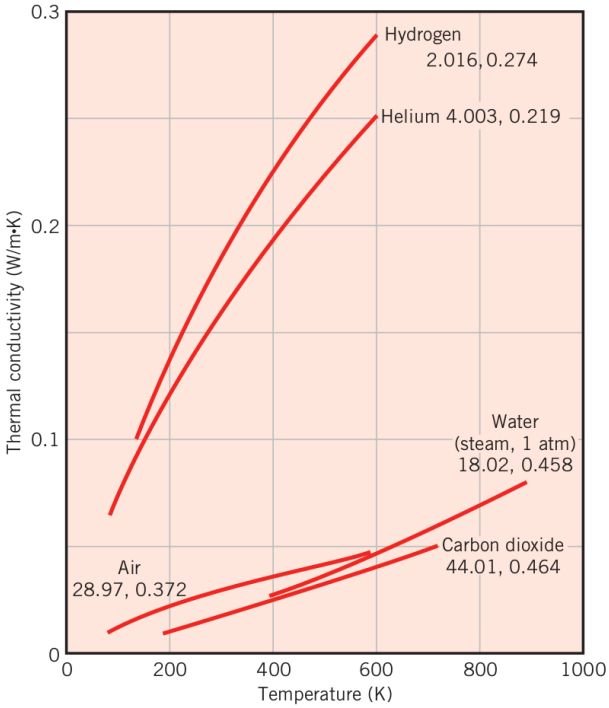
$\varepsilon = \frac{C_h(T_{h,i} - T_{h,o})}{C_{min}(T_{h,i} - T_{c,i})}$

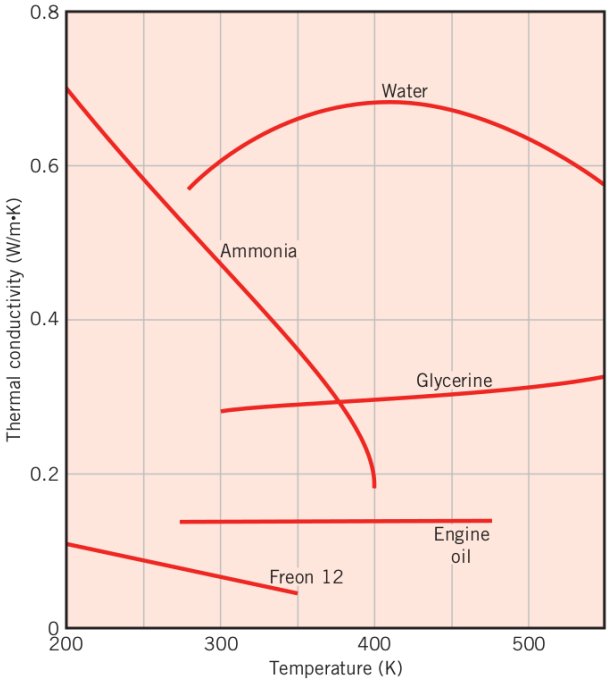
$\varepsilon = \frac{C_c(T_{c,o} - T_{c,i})}{C_{min}(T_{h,i} - T_{c,i})}$

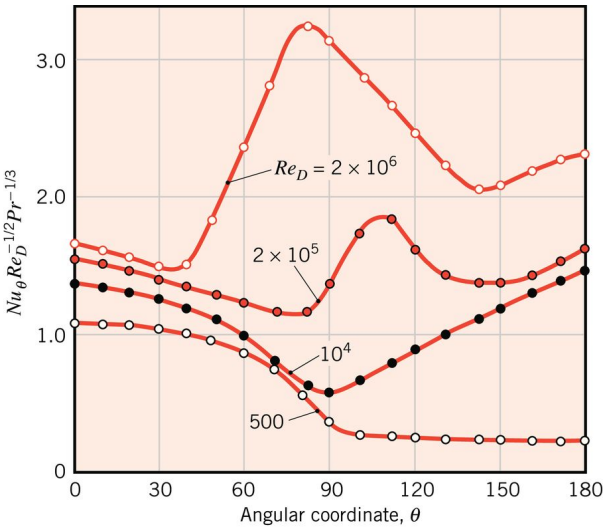
$q = \varepsilon C_{min}\left(T_{h,i} - T_{c,i}\right)$

$NTU = \frac{UA}{C_{min}}$

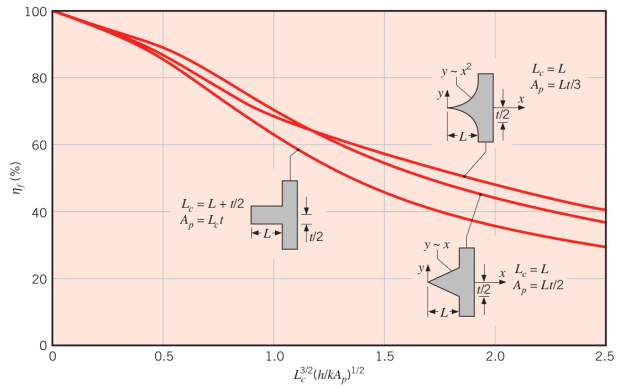

Thermal conductivity (W/m•K)
Temperature (K)
Silver
Copper
Gold
Aluminum
Aluminum alloy 2024
Tungsten
Platinum
Iron
Stainless steel, AISI 304
Aluminum oxide
Pyroceram
Fused quartz

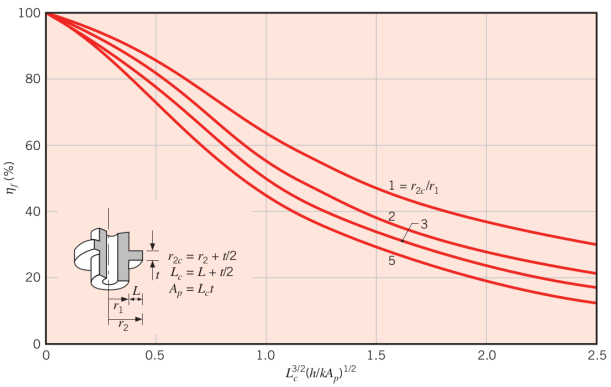

Thermal conductivity (W/m•K)
Temperature (K)
Hydrogen
2.016, 0.274
Helium 4.003, 0.219
Water
(steam, 1 atm)
18.02, 0.458
Carbon dioxide
44.01, 0.464
Air
28.97, 0.372


Thermal conductivity (W/m•K)
Temperature (K)
Water
Ammonia
Glycerine
Engine oil
Freon 12


$Nu_\theta Re_D^{-1/2} Pr^{-1/3}$
Angular coordinate, θ
$Re_D = 2 \times 10^6$
2×10^5
10^4
500

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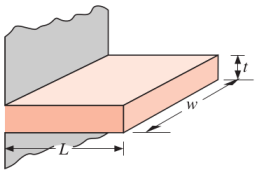
Straight Fins

Rectangular

$A_f = 2wL_c$

$L_c = L + (t/2)$

$A_p = tL$

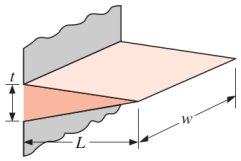


$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

Triangular

$A_f = 2w[L^2 + (t/2)^2]^{1/2}$

$A_p = (t/2)L$



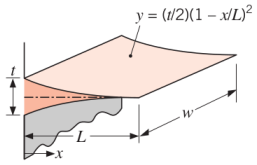
$$\eta_f = \frac{1}{mL}\frac{I_1(2mL)}{I_0(2mL)}$$

Parabolic

$A_f = w[C_1L + (L^2/t)\ln(t/L + C_1)]$

$C_1 = [1 + (t/L)^2]^{1/2}$

$A_p = (t/3)L$



$$\eta_f = \frac{2}{[4(mL)^2 + 1]^{1/2} + 1}$$

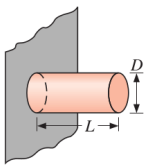
Pin Fins

Rectangular

$A_f = \pi DL_c$

$L_c = L + (D/4)$

$V = (\pi D^2/4)L$

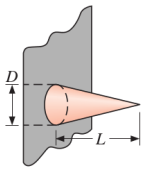


$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

Triangular

$A_f = \frac{\pi D}{2}[L^2 + (D/2)^2]^{1/2}$

$V = (\pi/12)D^2L$



$$\eta_f = \frac{2}{mL}\frac{I_2(2mL)}{I_1(2mL)}$$

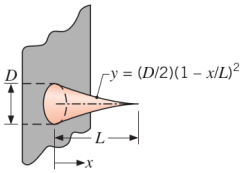
Parabolic

$A_f = \frac{\pi L^3}{8D}\{C_3C_4 - \frac{L}{2D}\ln[(2DC_4/L) + C_3]\}$

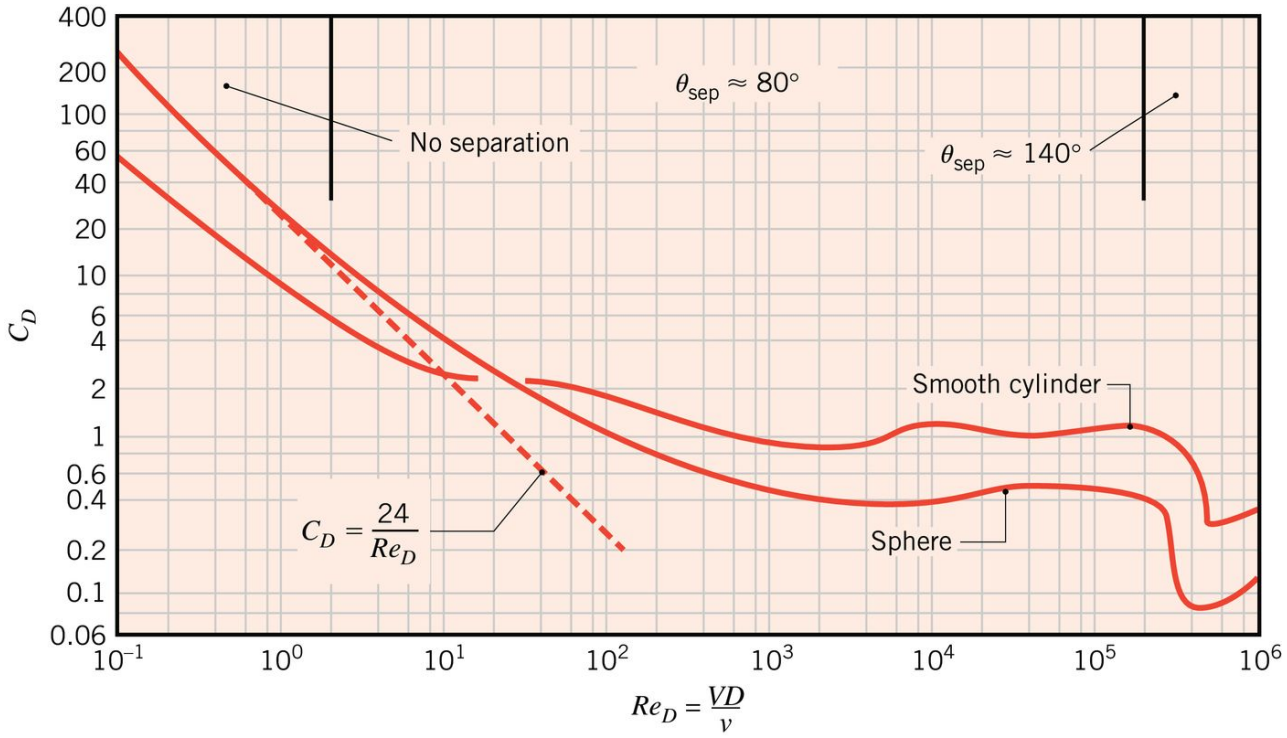
$C_3 = 1 + 2(D/L)^2$

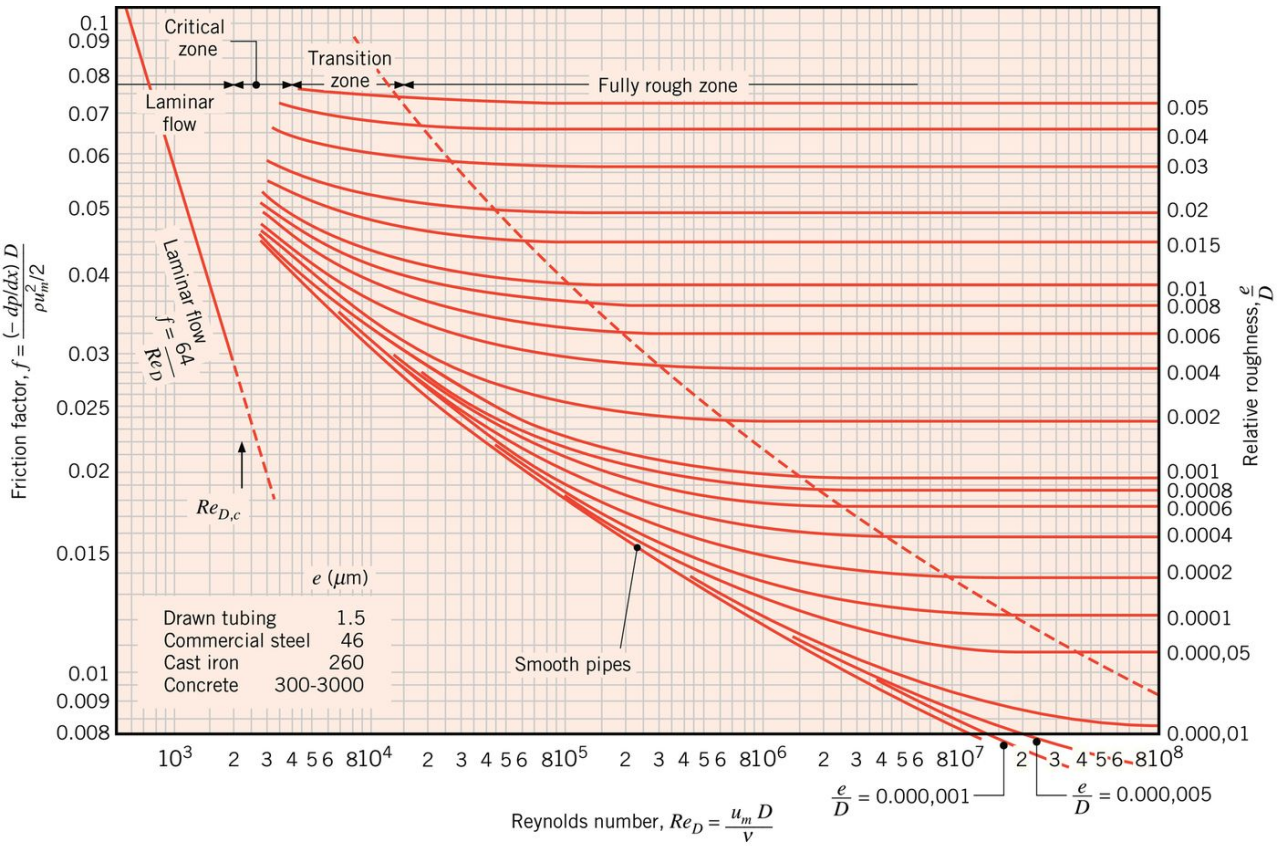
$C_4 = [1 + (D/L)^2]^{1/2}$

$V = (\pi/20)D^2L$



$$\eta_f = \frac{2}{[4/9(mL)^2 + 1]^{1/2} + 1}$$


$\theta_{sep} \approx 80°$
No separation
$\theta_{sep} \approx 140°$
Smooth cylinder
Sphere
$C_D = \frac{24}{Re_D}$
C_D
$Re_D = \frac{VD}{\nu}$

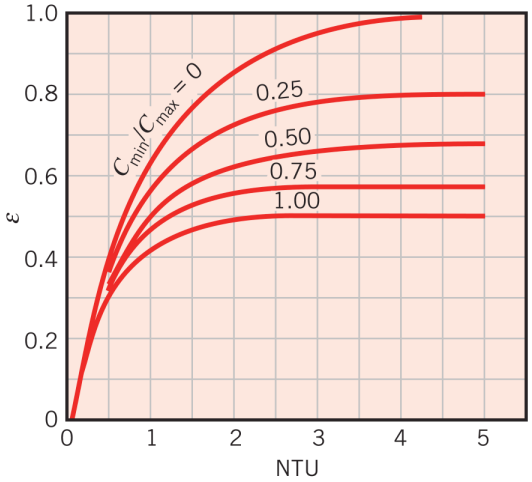

Critical zone
Transition zone
Fully rough zone
Laminar flow
Laminar flow $f = \frac{64}{Re_D}$
$Re_{D,c}$
e (μm)
Drawn tubing 1.5
Commercial steel 46
Cast iron 260
Concrete 300-3000
Smooth pipes
$\frac{e}{D}$ = 0.000,001
$\frac{e}{D}$ = 0.000,005
Friction factor, $f = \frac{(-dp/dx)D}{\rho u_m^2/2}$
Relative roughness, $\frac{e}{D}$
Reynolds number, $Re_D = \frac{u_m D}{\nu}$

Flow Arrangement	Relation
Parallel ow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 + C_r)]}{1 + C_r}$
Counterow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 - C_r)]}{1 - C_r \exp[-\text{NTU}(1 - C_r)]} \quad (C_r < 1)$
	$\varepsilon = \dfrac{\text{NTU}}{1 + \text{NTU}} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, . . . tube passes)	$\varepsilon_1 = 2\left\{1 + C_r + (1 + C_r^2)^{1/2} \times \dfrac{1 + \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}\right\}^{-1}$
n shell passes ($2n$, $4n$, . . . tube passes)	$\varepsilon = \left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - 1\right]\left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - C_r\right]^{-1}$
Cross-ow (single pass)	
Both fluids unmixed	$\varepsilon = 1 - \exp\left[\left(\dfrac{1}{C_r}\right)(\text{NTU})^{0.22}\{\exp[-C_r(\text{NTU})^{0.78}] - 1\}\right]$
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\varepsilon = \left(\dfrac{1}{C_r}\right)(1 - \exp\{-C_r[1 - \exp(-\text{NTU})]\})$
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\varepsilon = 1 - \exp(-C_r^{-1}\{1 - \exp[-C_r(\text{NTU})]\})$
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-\text{NTU})$

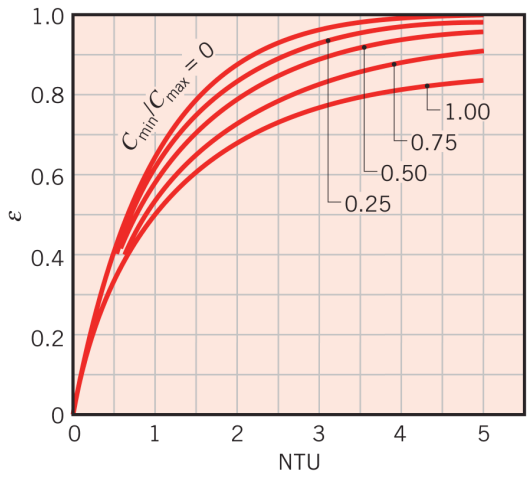
Flow Arrangement	Relation
Parallel ow	$\text{NTU} = -\dfrac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$
Counterow	$\text{NTU} = \dfrac{1}{C_r - 1}\ln\left(\dfrac{\varepsilon - 1}{\varepsilon C_r - 1}\right) \quad (C_r < 1)$
	$\text{NTU} = \dfrac{\varepsilon}{1 - \varepsilon} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, . . . tube passes)	$(\text{NTU})_1 = -(1 + C_r^2)^{-1/2}\ln\left(\dfrac{E - 1}{E + 1}\right)$ $E = \dfrac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$
n shell passes ($2n$, $4n$, . . . tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \dfrac{F - 1}{F - C_r} \quad F = \left(\dfrac{\varepsilon C_r - 1}{\varepsilon - 1}\right)^{1/n} \quad \text{NTU} = n(\text{NTU})_1$
Cross-ow (single pass)	
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\text{NTU} = -\ln\left[1 + \left(\dfrac{1}{C_r}\right)\ln(1 - \varepsilon C_r)\right]$
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\text{NTU} = -\left(\dfrac{1}{C_r}\right)\ln[C_r \ln(1 - \varepsilon) + 1]$
All exchangers ($C_r = 0$)	$\text{NTU} = -\ln(1 - \varepsilon)$

ENGT 320 Applied Thermal Systems
Quiz 4 Formula Sheet

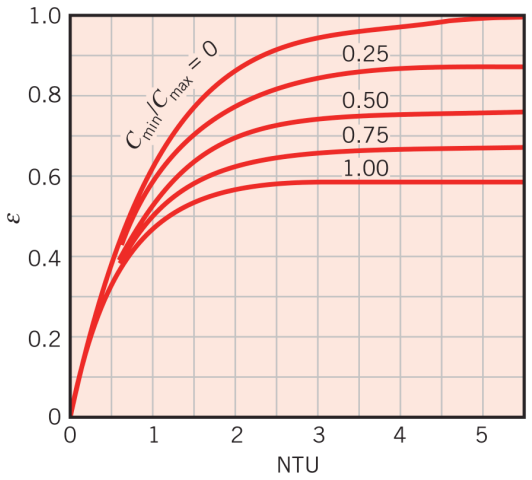
Parallel Flow



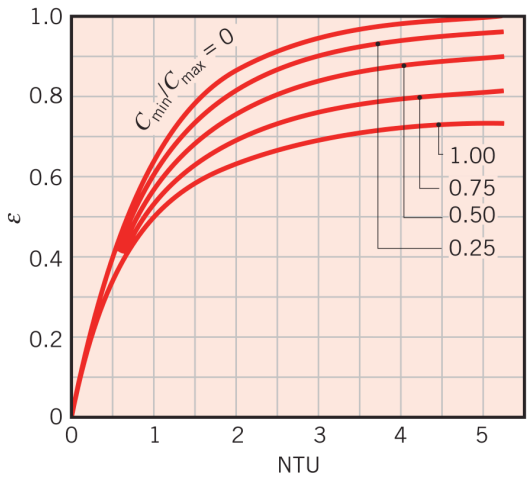
Counter Flow



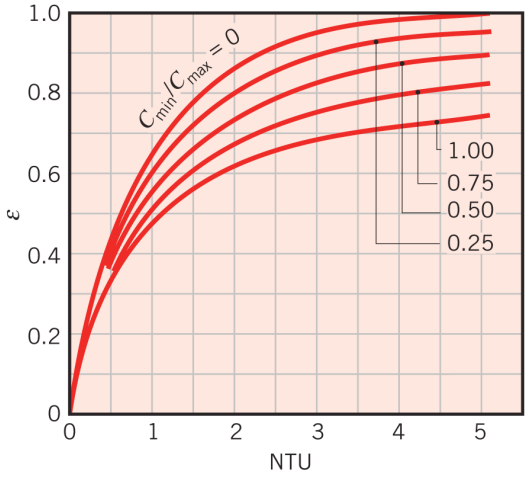
Shell and Tube - One Shell



Shell and Tube - 2 Shells



Cross-flow Both Unmixed



Cross-flow One Mixed/One Unmixed

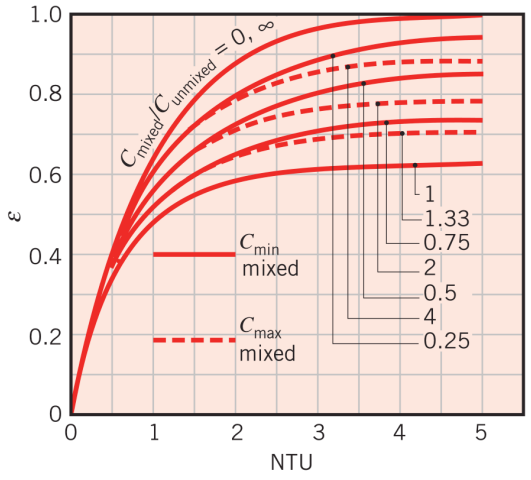


Table 1: Contact Resistance for vacuum interfaces, $R''_{t,c} \times 10^4 \left(\frac{m^2 K}{W}\right)$

Material	$100\,(kPa)$	$10,000\,(kPa)$
Stainless Steel	6-25	0.7-4.0
Copper	1-10	0.1-0.5
Magnesium	1.5-3.5	0.2-0.4
Aluminum	1.5-5.0	0.2-0.4

Table 2: Contact Resistance for aluminum ($10\mu m$ surface roughness, $10kPa$ contact pressure)

Interfacial Fluid	$R''_{t,c} \times 10^4 \left(\frac{m^2 K}{W}\right)$
Air	2.75
Helium	1.05
Hydrogen	0.720
Silicone Oil	0.525
Glycerine	0.265

Table 3: Fin Tip Conditions

Case	Tip Condition	Temperature Distribution $\frac{\theta}{\theta_b}$	Fin Heat Transfer Rate q_f
A	Convection	$\frac{\cosh m(L-x)+\left(\frac{h}{mk}\right)\sinh m(L-x)}{\cosh mL+\left(\frac{h}{mK}\right)\sinh mL}$	$M\frac{\sinh mL+\left(\frac{h}{mk}\right)\cosh mL}{\cosh mL+\left(\frac{h}{mk}\right)\sinh mL}$
B	Adiabatic	$\frac{\cosh m(L-x)}{\cosh mL}$	$M\tanh mL$
C	Constant Temperature	$\frac{\left(\frac{\theta_L}{\theta_b}\right)\sinh mx+\sinh m(L-x)}{\sinh mL}$	$M\frac{\cosh mL-\frac{\theta_L}{\theta_b}}{\sinh mL}$
D	Infinite Fin ($L\to\infty$)	e^{-mx}	M

Table 4: Energy Balance Method Case Summary

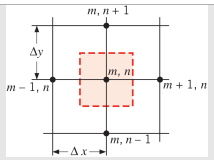
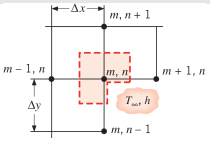
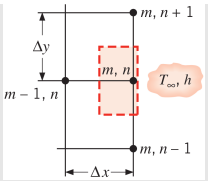
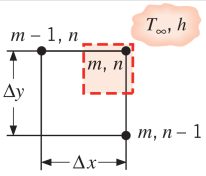
Case	Diagram	Equation
1		$T_{m,n+1} + T_{m,n-1} + T_{m+1,n} + T_{m-1,n} - 4T_{m,n} = 0$
2		$2\left(T_{m-1,n} - T_{m,n+1}\right) + \left(T_{m+1,n} + T_{m,n-1}\right) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$
3		$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(2 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$
4		$(T_{m,n-1} + T_{m-1,n}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(1 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$

Table 5: Cylinder In Cross Flow

Re_D	C	m
$0.4 - 4$	0.989	0.330
$4 - 40$	0.911	0.385
$40 - 4000$	0.683	0.466
$4000 - 40,000$	0.193	0.618
$40,000 - 400,000$	0.027	0.805

Table 6: Various Geometries In Cross Flow

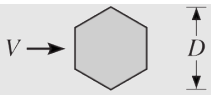
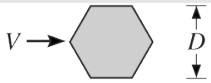
Geometry		Re_D	C	m
		$6000 - 60,000$	0.304	0.59
		$5000 - 60,000$	0.158	0.66
		$5200 - 20,400$ $20,400 - 105,000$	0.164 0.039	0.638 0.78
		$4500 - 90,700$	0.150	0.638
	Front Back	$10,000 - 50,000$ $7,000 - 80,000$	0.667 0.191	0.500 0.667

Table 7: Noncircular Tubes

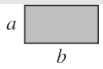
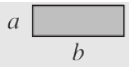
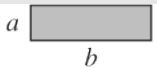
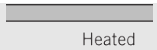
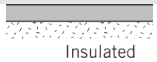
Cross Section	$\frac{b}{a}$	Uniform q_s''	Uniform T_s	fRe_{D_h}
		4.36	3.66	64
	1.0	3.61	2.98	57
	1.43	3.73	3.08	59
	2.0	4.12	3.39	62
	3.0	4.79	3.96	69
	4.0	5.33	4.44	73
	∞	8.23	7.54	96
	∞	5.39	4.86	96
		3.11	2.49	53

Table 8: Fouling Factors

Fluid	$R_f^{''}\left(m^2K/W\right)$
Seawater and treated boiler feedwater (below $50°C$)	0.001
Seawater and treated boiler feedwater (above $50°C$)	0.002
River water (below $50°C$)	0.0002 - 0.001
Fuel Oil	0.0009
Refrigerating Liquids	0.0002
Steam (nonoil bearing)	0.0001